

UAV-Based Coastal Marine Debris Segmentation via Dynamic Feature Fusion

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Abstract—Automated monitoring of beach debris using UAV imagery is promising for pollution management but faces visual challenges including complex background textures and partial burial. This study proposes an adaptive instance segmentation model based on YOLOv26n-seg to extract masks of marine debris on beach. The architecture integrates a dynamic feature fusion structure C3kDFF to suppress background noise, a context anchor attention module CPCAA to capture spatial dependencies for thin objects, and a dynamic multi scale sequence fusion module DMSSF, to preserve structural continuity. Experimental results show that the model achieves an mAP_{50} of 58.7% representing a 5.1% improvement over the baseline. Visual analysis shows that the model reduces missed detections in environments with strong illumination or seagrass wrack. By improving detection robustness the framework provides a reliable tool for quantifying marine debris location and area supporting effective coastal preservation strategies.

Index Terms—UAV remote sensing, instance segmentation, marine debris, deep learning, beach monitoring

I. INTRODUCTION

Marine debris is recognized as a major global environmental threat [1]. Also, beach shorelines serve as marine debris sink [1]. These sinks pose risks to marine biodiversity and causes economic losses in tourism [3]. Since the environmental impact of debris is related to its material and geometric properties, high resolution and low cost UAV remote sensing has become an essential tool. Automated monitoring methods combining UAV with computer vision have been increasingly applied to this field [2].

However, natural beach environments present visual challenges such as complex textures and partial debris burial. Existing segmentation methods based on static feature fusion often fail to produce precise masks in coastal conditions. Therefore, we propose an architecture featuring dynamic selection and context aware attention. This study provides an efficient model for marine debris monitoring.

II. METHODOLOGY

The methodology focuses on optimizing the feature extraction and fusion capabilities of the baseline architecture.

The overall framework illustrates the UAV data acquisition and segmentation process as Fig. 1.

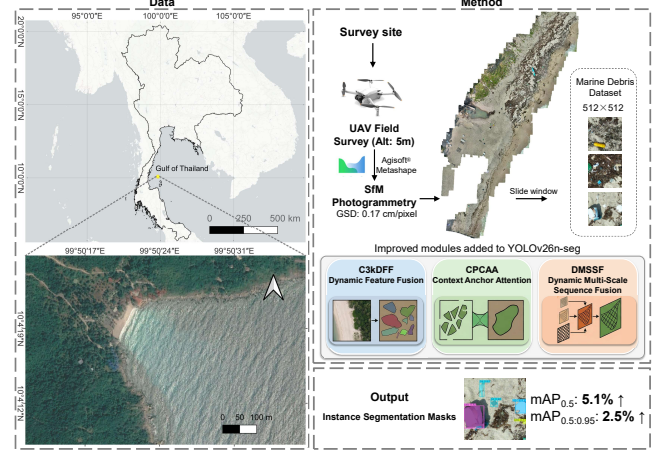


Fig. 1 Overall framework from data acquisition, SfM Photogrammetry and instance segmentation process.

A. Dynamic Feature Fusion Structure C3kDFF

C3kDFF addresses the limitation of static residual connections in standard feature extractors. Small objects like bottle caps often share visual similarities with the background. Inspired by the dynamic selection mechanism [4], we design the C3kDFF structure using a channel and space dual domain selection process. It applies global average pooling and a gating mechanism to identify important channels while computing spatial attention weights to highlight debris regions. This allows the model to adapt to background conditions.

B. Context Anchor Attention Module CPCAA

Beach litter often includes thin and irregular objects that are difficult to locate using local textures alone. We integrate the context anchor attention mechanism [5] into the structure to improve spatial awareness. CPCAA uses horizontal

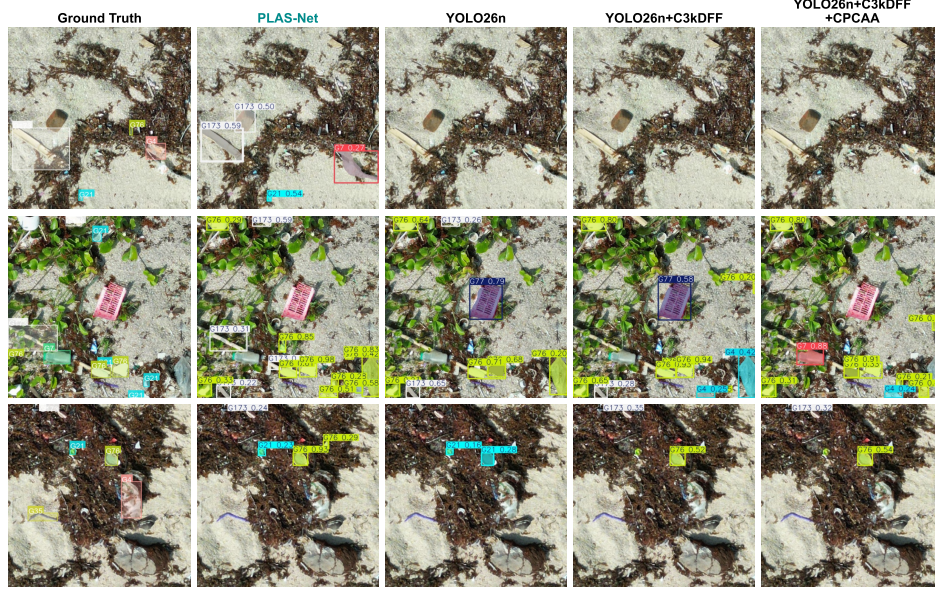


Fig. 2 Ablation Experiment visual results (Ground Sample Distance (GSD): 0.17 cm/px, crop area: $\sim 0.87 \times 0.87$ m²). Colors represent EU Joint List marine debris categories (e.g., G7: plastic bottles ≤ 0.5 L, G4: small plastic bags).

and vertical strip convolutions to form directional receptive fields. This captures long distance spatial dependencies for elongated items while maintaining a low computational cost.

C. Dynamic Multi scale Sequence Fusion Module DMSSF

Standard feature pyramids often fail to capture semantic consistency across different layers. We use the DMSSF module to model multi scale features as a sequence, utilizing a dynamic upsampling mechanism DySample [6] to resize feature maps. It then builds a three dimensional feature tensor and uses convolutions to extract cross scale associations which ensures the structural continuity of multi scale targets.

III. RESULTS

A. Ablation Study

Ablation experiments demonstrate that the baseline mAP_{50} of 53.6% increases to 55.5% with C3kDFF, and ultimately reaches 58.7% after integrating CPCAA and DMSSF. Visual analysis in Fig. 2 shows these modules significantly reduce false negatives and misidentifications in cluttered environments (e.g., strong illumination, white distractors, and flocculent seagrass wrack). Although the proposed model occasionally generates false positives in clean sand, prioritizing the discovery and accurate boundary delineation of actual pollutants outweighs this minor trade-off, especially given the long-tail distribution of marine debris.

B. Comparison with Mainstream Models

The proposed model is compared with several instance segmentation models. Our model achieves a higher mAP_{50} of 58.7% while maintaining a processing speed of 145 FPS.

The lightweight parameter count of 2.73M makes it suitable for deployment on UAV platforms.

IV. CONCLUSION

This study proposed an adaptive instance segmentation framework to address the visual complexities of coastal marine debris monitoring. Integrating dynamic feature fusion and context-aware attention mechanisms, the model achieves a 5.1% improvement in mAP_{50} over the baseline. This result ensures reliable pixel-level quantification of debris size and spatial distribution. Furthermore, with an lightweight design, the proposed model facilitates real-time inference on UAV edge computing platforms during field surveys. Future work will involve expanding the dataset to diverse coastal environments, and developing temporal tracking methods to monitor debris accumulation dynamics over time.

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